THE LEADING COEFFICIENT OF THE LENGTH OF THE QUOTIENT BY POWERS OF A PARAMETER IDEAL IS BOUNDED BY THE LENGTH OF THE QUOTIENT BY THE IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author refers to the earlier statement as 'Lemma' without a number in one sentence and uses the abbreviation 'Lemma' for 'the lemma' at the end; both are reproduced unchanged. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The Hilbert function of a parameter ideal is a numerical polynomial of degree equal to the dimension. invoked at proof lines 38270-38273. Makes the leading coefficient e_I/d! meaningful and fixes the degree.
- Surjectivity of the monomial map onto the graded pieces I^n/I^{n+1}. used at proof lines 38276-38283. Produces the length inequality between consecutive quotients.
- Comparison of leading coefficients of two numerical polynomials of the same degree. used at proof lines 38289-38293. Converts the inequality between polynomials into the numerical bound on e_I.

Source reference. Original lemma 00KQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 760–760. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00K8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 742–742. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 37979: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37979–37979. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension d, let $g_1, \ldots, g_d$ be a system of parameters, and let $I = (g_1, \ldots, g_d)$. If $e_I/d!$ is the leading coefficient of the numerical polynomial $n \mapsto \text{length}_R(R/I^{n+1})$, then $e_I \leq \text{length}_R(R/I)$.*

Proof. The function is a numerical polynomial by Algebra, Proposition 00K8. It has degree d by Algebra, Proposition 00KQ. If $d = 0$, then the result is trivial. If $d = 1$, then the result is Lemma 37979. To prove it in general, observe that there is a surjection

$$\bigoplus\nolimits_{i_1,\ldots,i_d \geq 0,\ \sum i_j = n} R/I \longrightarrow I^n/I^{n+1}$$

sending the basis element corresponding to $i_1, \ldots, i_d$ to the class of $g_1^{i_1} \ldots g_d^{i_d}$ in I^n/I^{n+1}. Thus we see that

$$\text{length}_R(R/I^{n+1}) - \text{length}_R(R/I^n) \leq \text{length}_R(R/I)\binom{n+d-1}{d-1}$$

Since $d \geq 2$ the numerical polynomial on the left has degree $d-1$ with leading coefficient $e_I/(d-1)!$. The polynomial on the right has degree $d-1$ and its leading coefficient is $\text{length}_R(R/I)/(d-1)!$. This proves the lemma. □